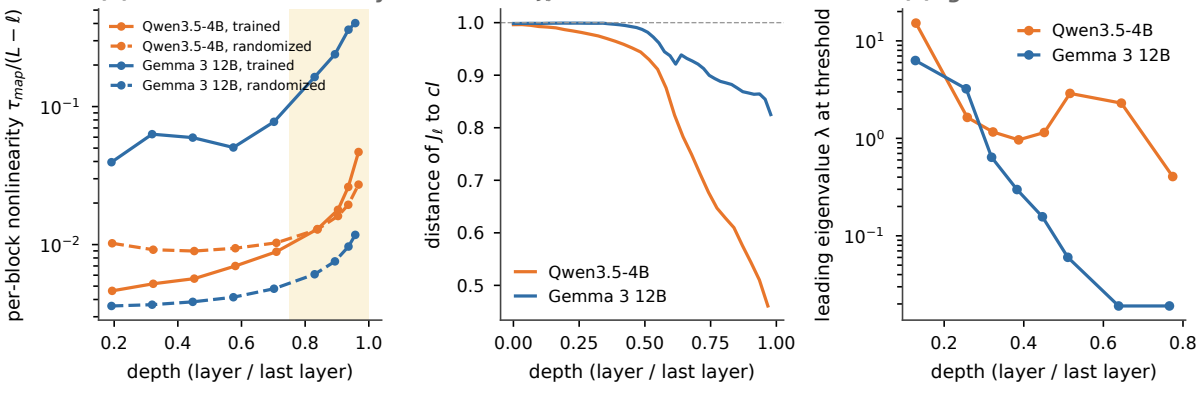
(a) Per-block nonlinearity
Qwen3.5-4B, trained
Qwen3.5-4B, randomized
Gemma 3 12B, trained
Gemma 3 12B, randomized
per-block nonlinearity $\tau_{map}/(L-\ell)$
10^{-1}
10^{-2}
0.2
0.4
0.6
0.8
1.0
depth (layer / last layer)
(b) $J_\ell \to I$
distance of J_ℓ to cI
1.0
0.9
0.8
0.7
0.6
0.5
Qwen3.5-4B
Gemma 3 12B
0.00
0.25
0.50
0.75
1.00
depth (layer / last layer)
(c) Ignition curvature
leading eigenvalue λ at threshold
Qwen3.5-4B
Gemma 3 12B
10^{1}
10^{0}
10^{-1}
0.2
0.4
0.6
0.8
depth (layer / last layer)